

Linear models

Readings for today

- Chapter 3: Linear regression. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). An introduction to statistical learning: with applications in R (Vol. 6). New York: Springer.

Topics

1. Ordinary least squares regression
2. Polynomial models
3. Isomorphisms with other statistical tests

Ordinary least squares regression

Structure of a linear regression model

Fundamental form:

$$Y = f(X) + \epsilon$$
$$= \underbrace{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}_{\text{slope} * x + \text{intercept}} + \epsilon$$

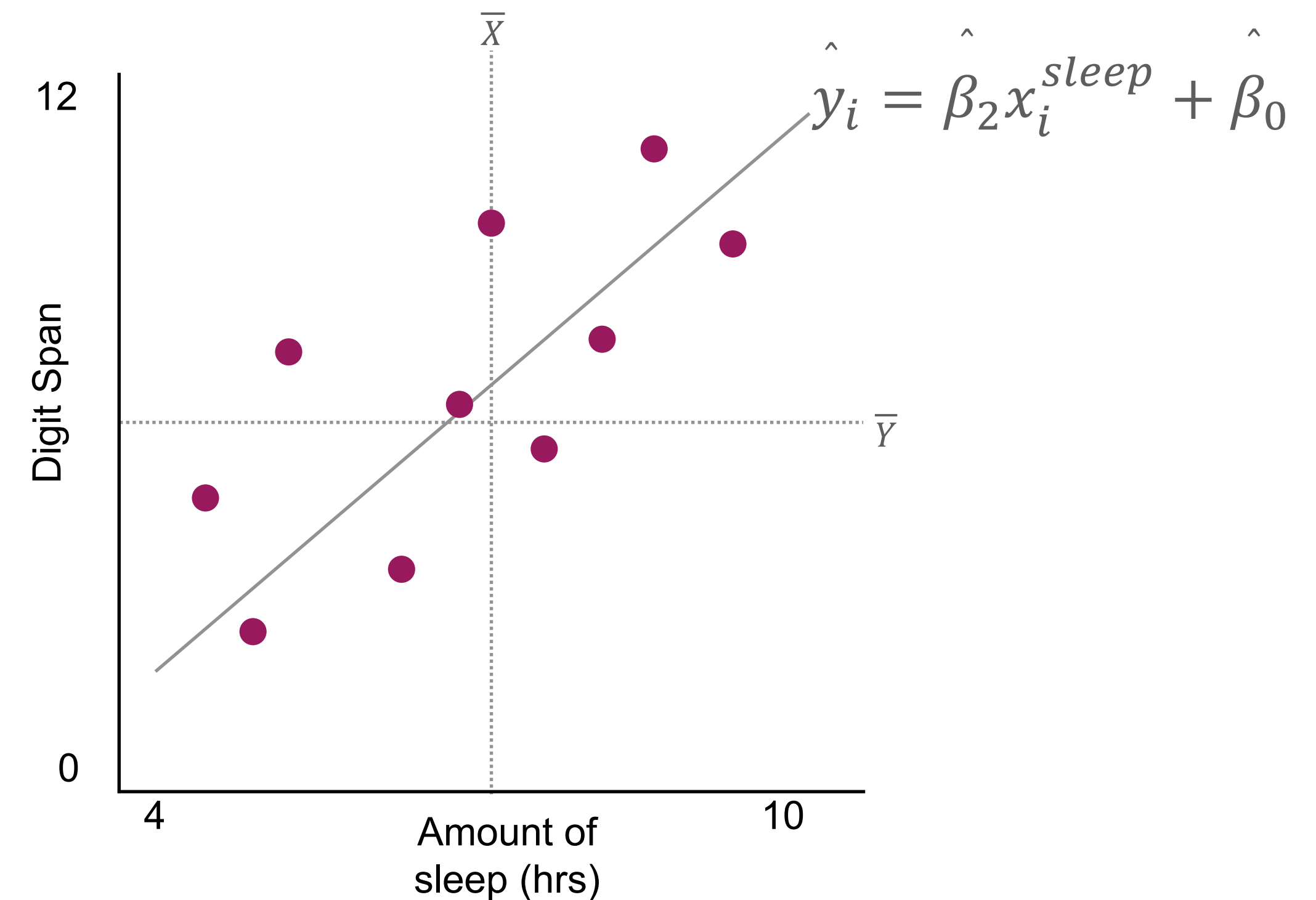
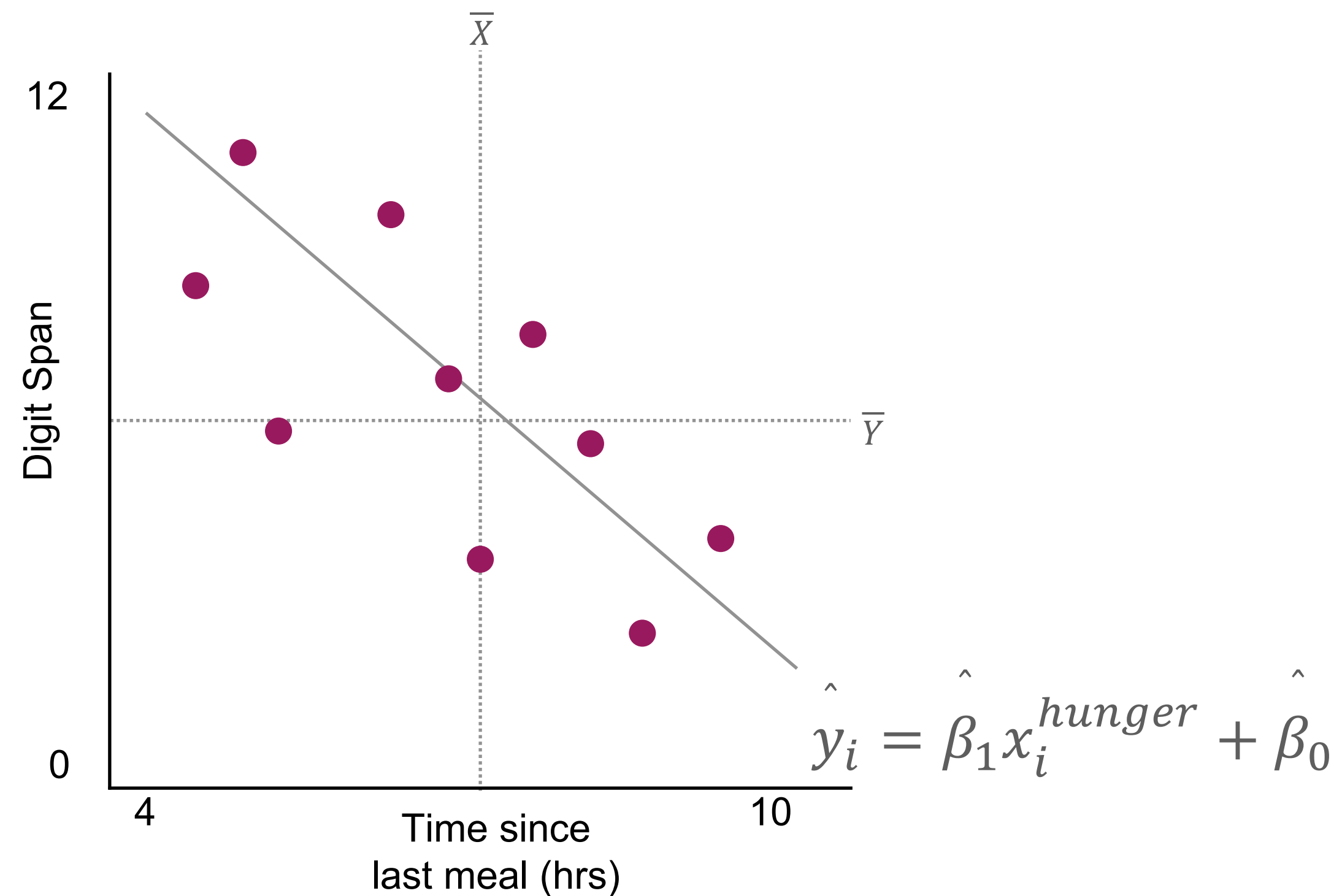
Necessary Assumptions to compute Ordinary Least Squares (OLS):

1. **Linear model specification:** predictors enter linearly in the parameters ($Y = X\beta + \epsilon$)
2. **Enough observations** ($n \geq p$): otherwise the columns of X cannot be linearly independent.
3. **No perfect multicollinearity:** No predictor is an exact linear combination of others.

+ Residuals of Y are normally distributed

+ For every value x_i there is only one y_i

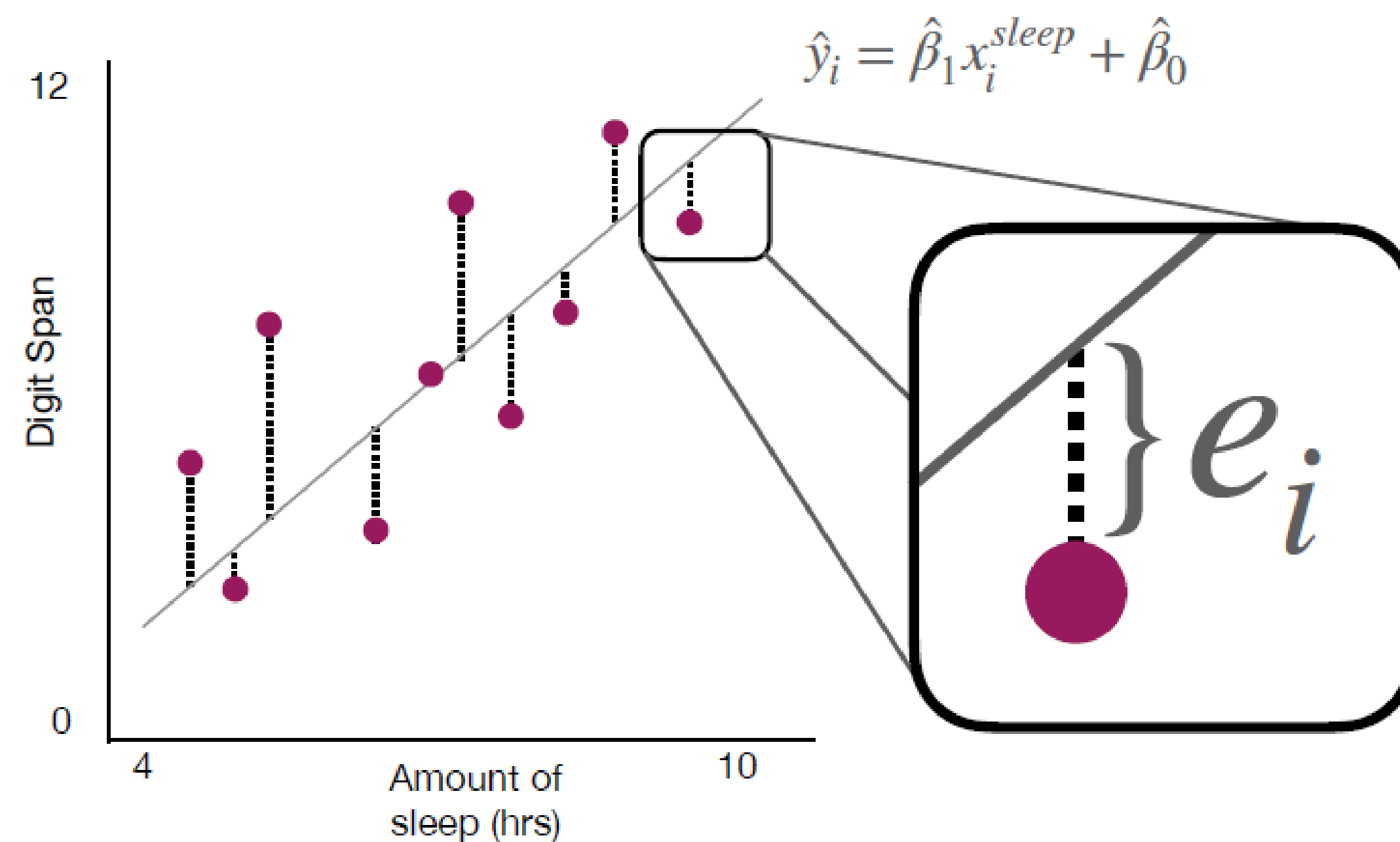
Structure of a linear regression model



Model: $\hat{y}_i = \hat{\beta}_1 x_i^{hunger} + \hat{\beta}_2 x_i^{sleep} + \hat{\beta}_0$

Ordinary least squares (OLS)

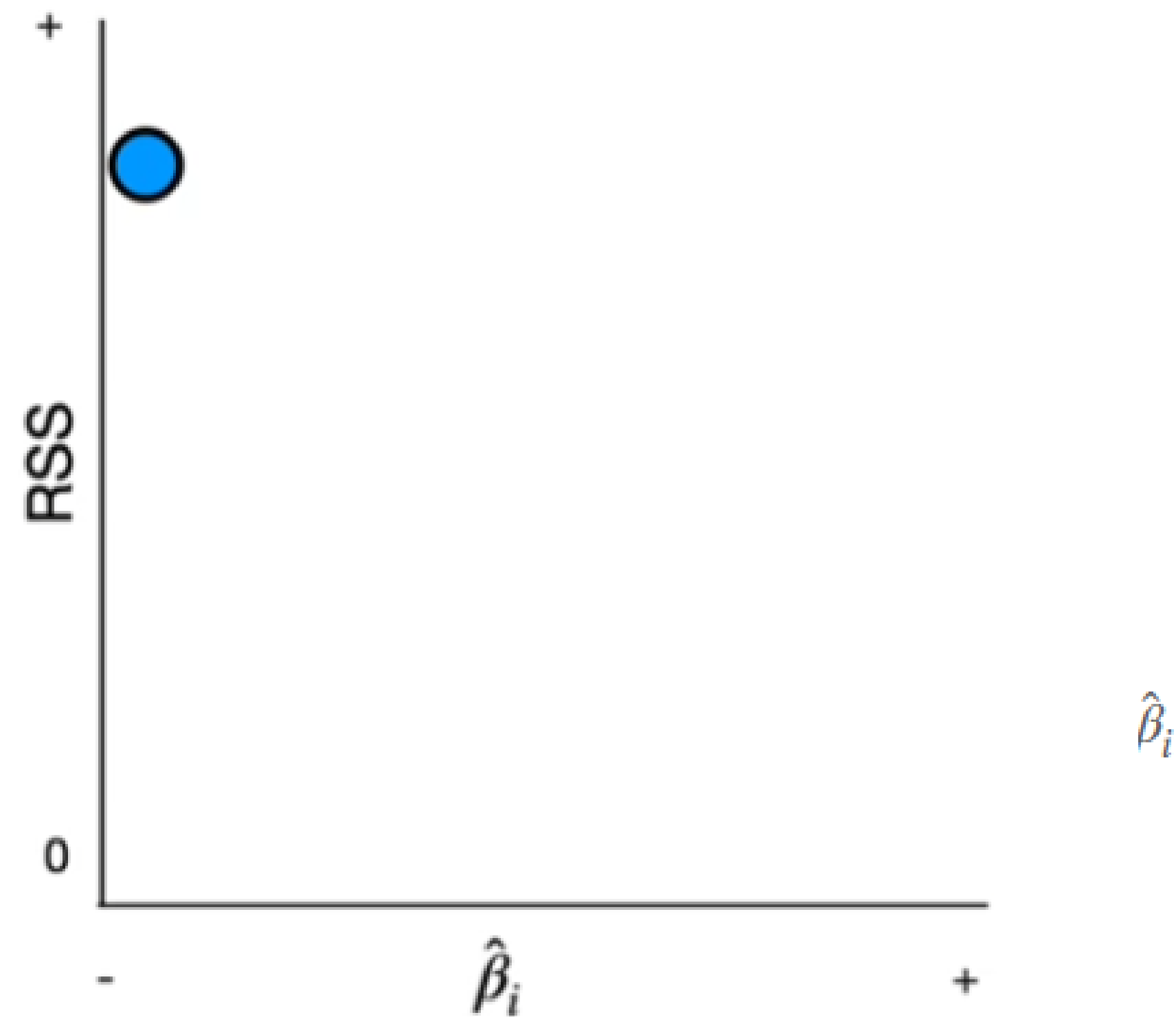
Error function: $e_i = y_i - \hat{y}_i$



Residual Sums of Squares (RSS):

$$\begin{aligned} RSS &= e_1^2 + \dots + e_n^2 \\ &= (y_1 - \hat{\beta}_1 x_1)^2 \dots (y_n - \hat{\beta}_1 x_n)^2 \\ &= \sum_{i=1}^n (y_i - \hat{\beta}_1 x_i)^2 \end{aligned}$$

Convexity & optimal solution



Ordinary Least Squares (OLS):

$$\begin{aligned}\hat{\beta}_0 &= E[Y] - \hat{\beta}_1 E[X] \\ &= \bar{y} - \hat{\beta}_1 \bar{x}\end{aligned}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Confidence of estimates

Types of error: $Y = \underbrace{f(X)}_{\text{reducible}} + \underbrace{\epsilon}_{\text{irreducible}}$

1. Residual square error: $RSE = \sigma_{model}^2 = \sqrt{\frac{RSS}{n-p-1}}$

2. Standard error of estimate: $SE(\hat{\beta}_i) = \sigma_{model}^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right)$

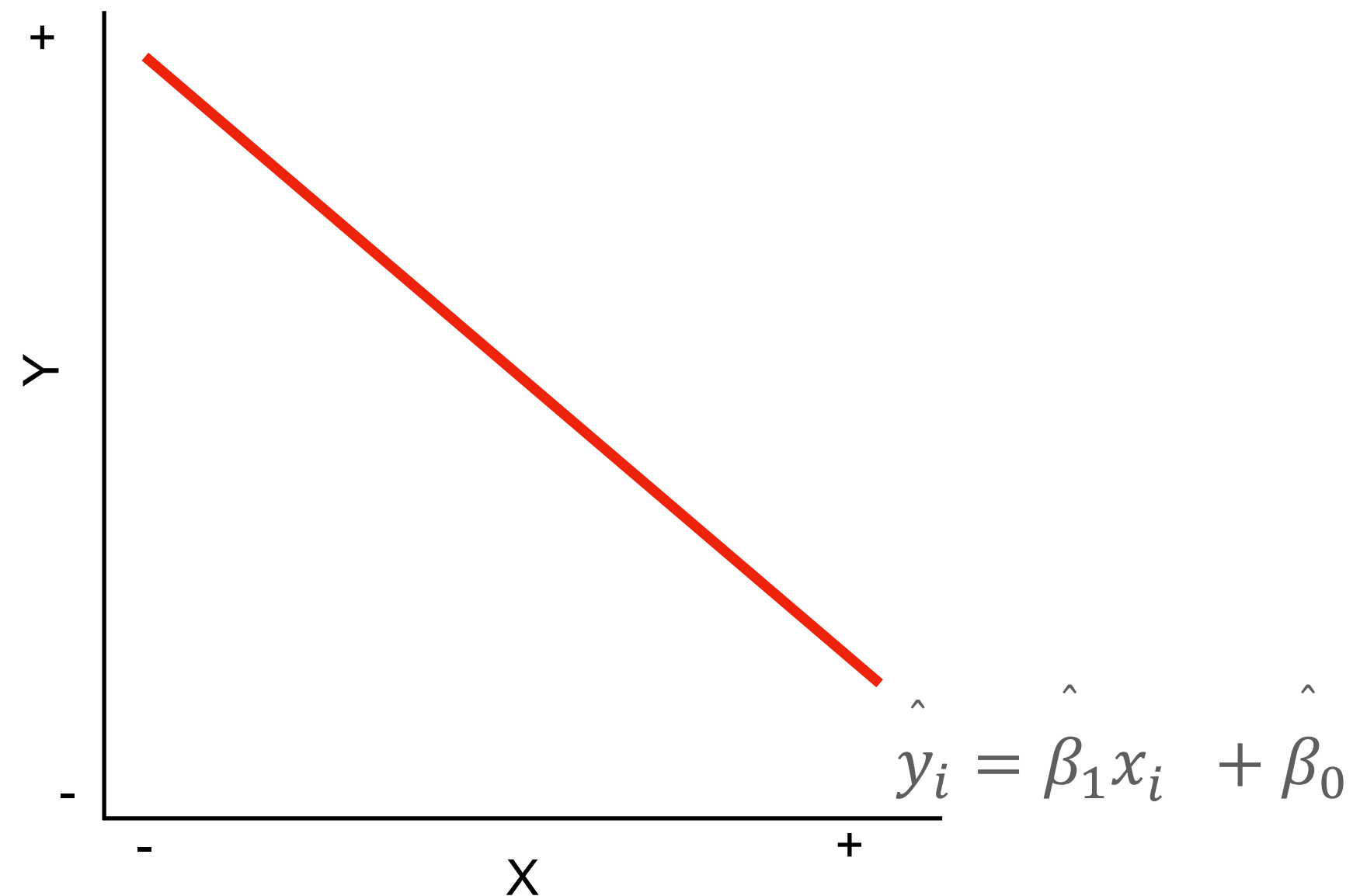
$$= \frac{\sigma_{model}^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Polynomial models

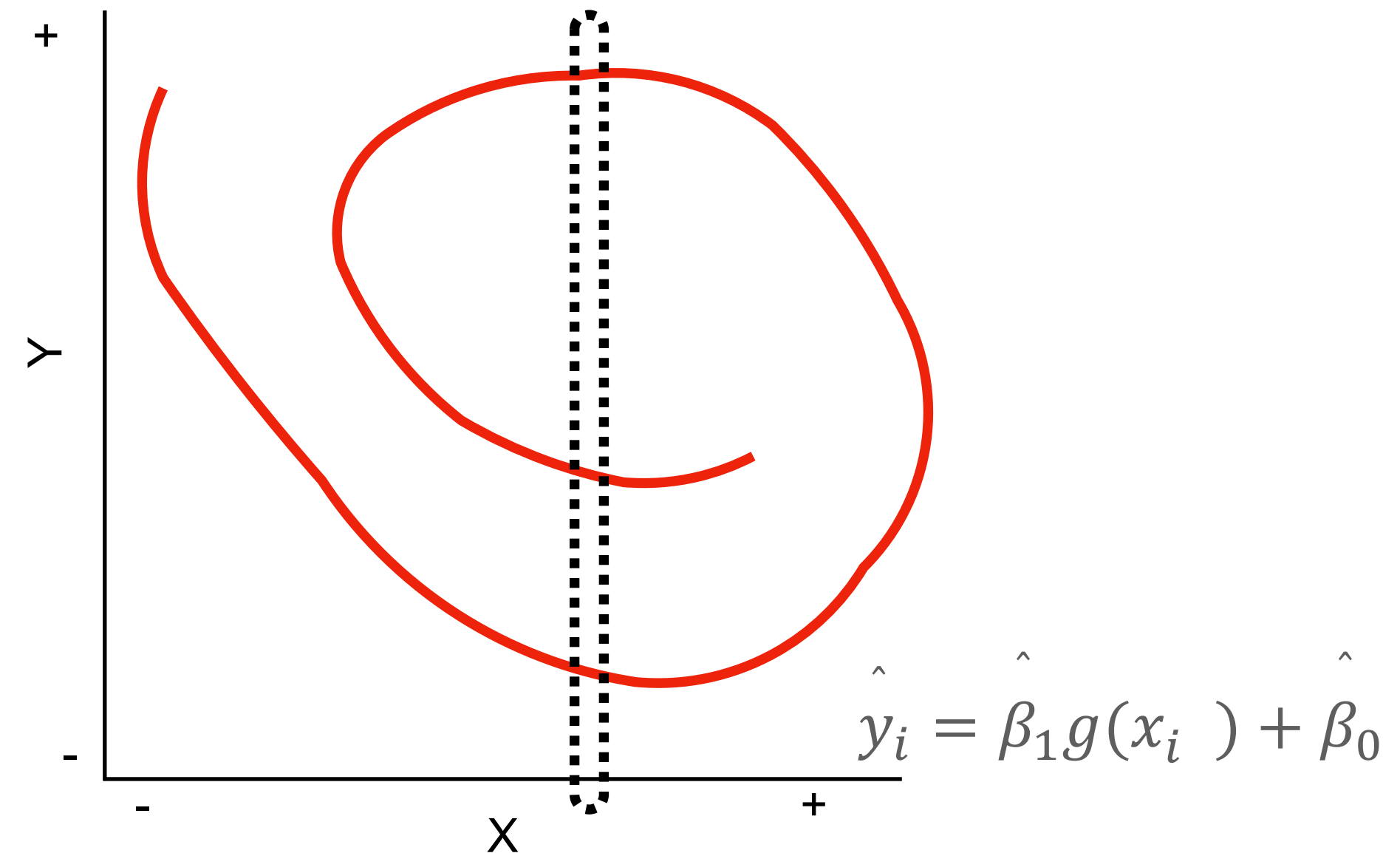
OLS solution works for all linear terms

- Linear system:
- Successive effects are *additive*.
 - For every unique x_i there exists *only one* possible y_i

Linear

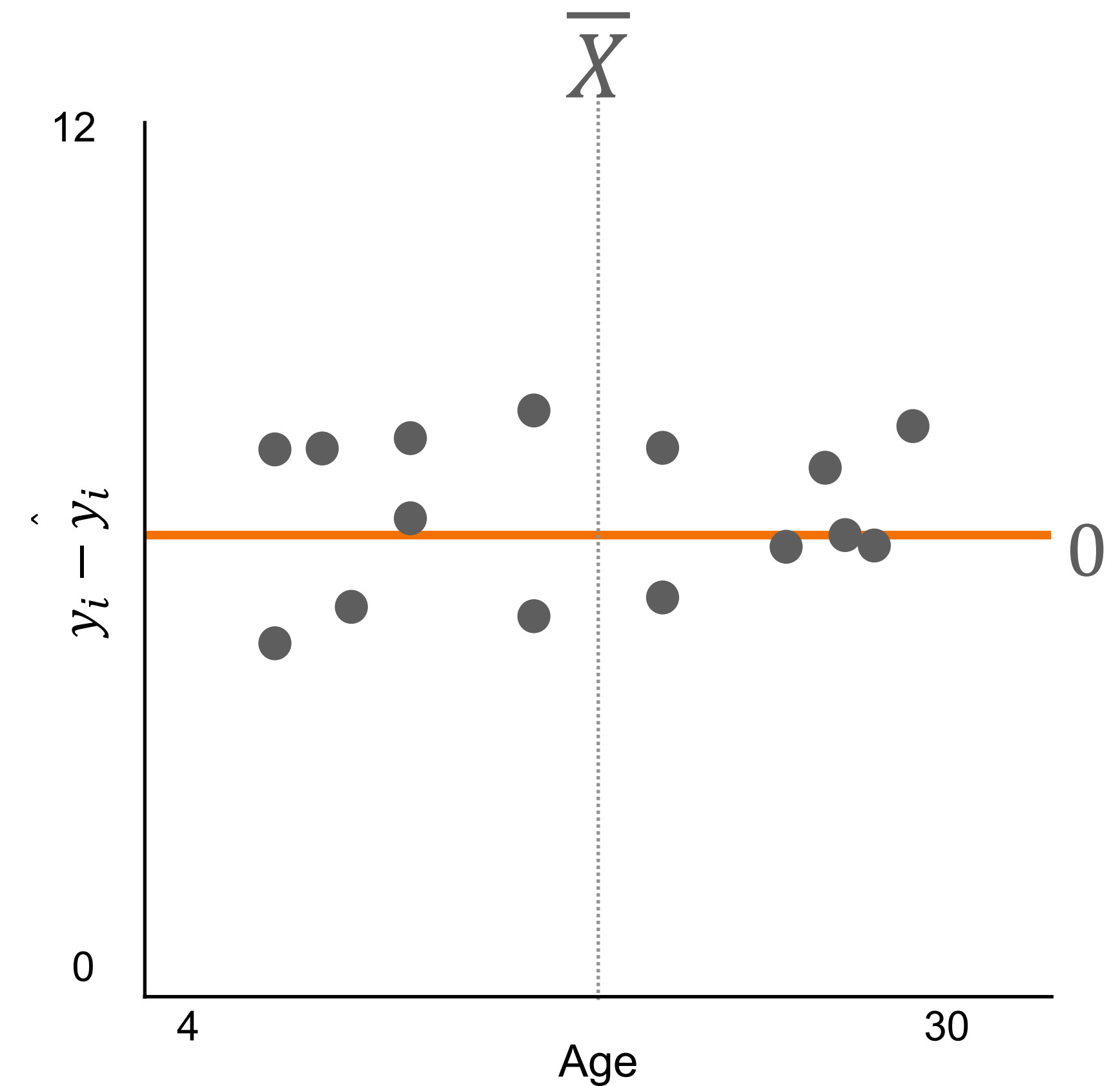
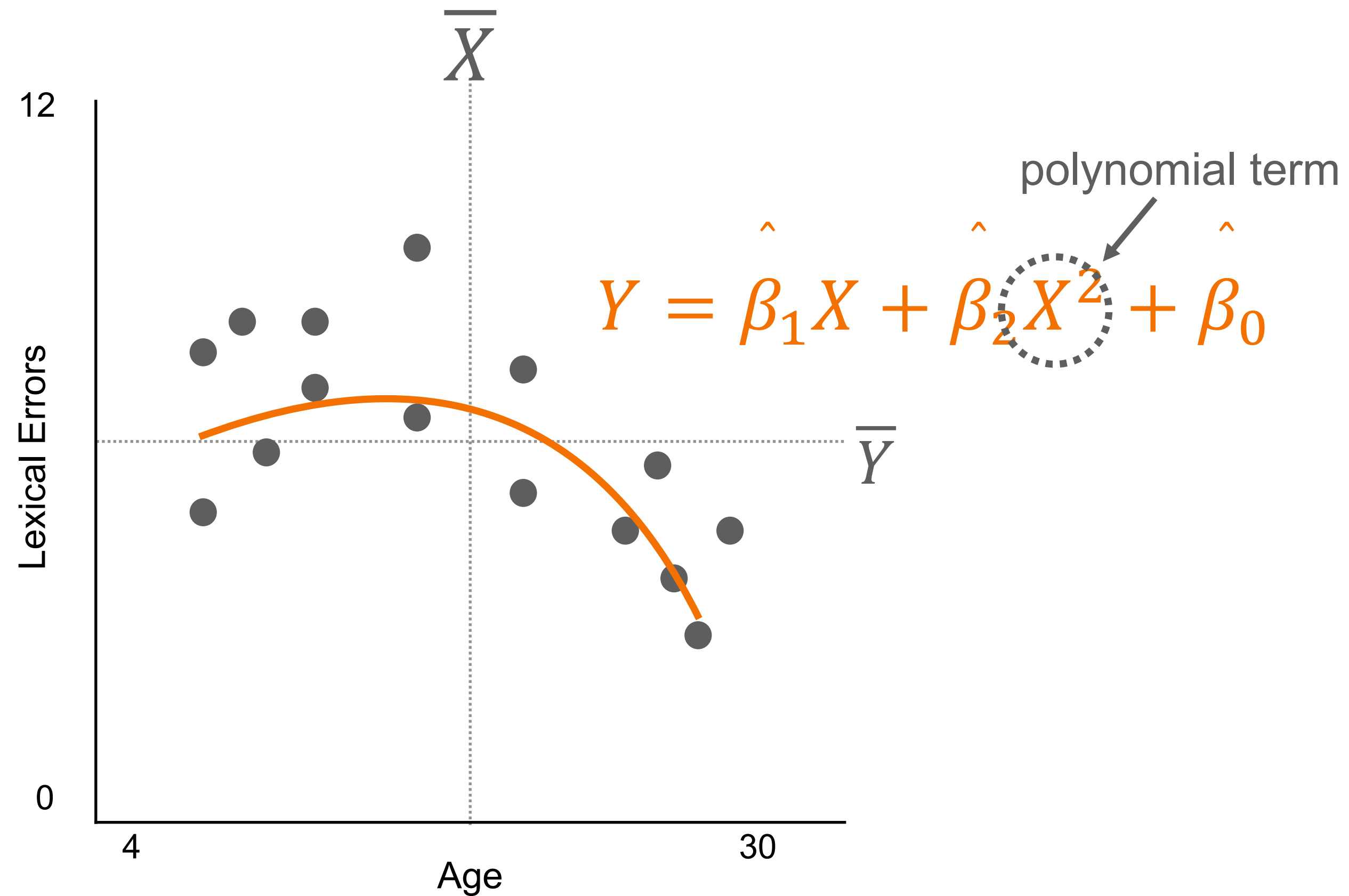


Non-linear



Polynomial regression

Example

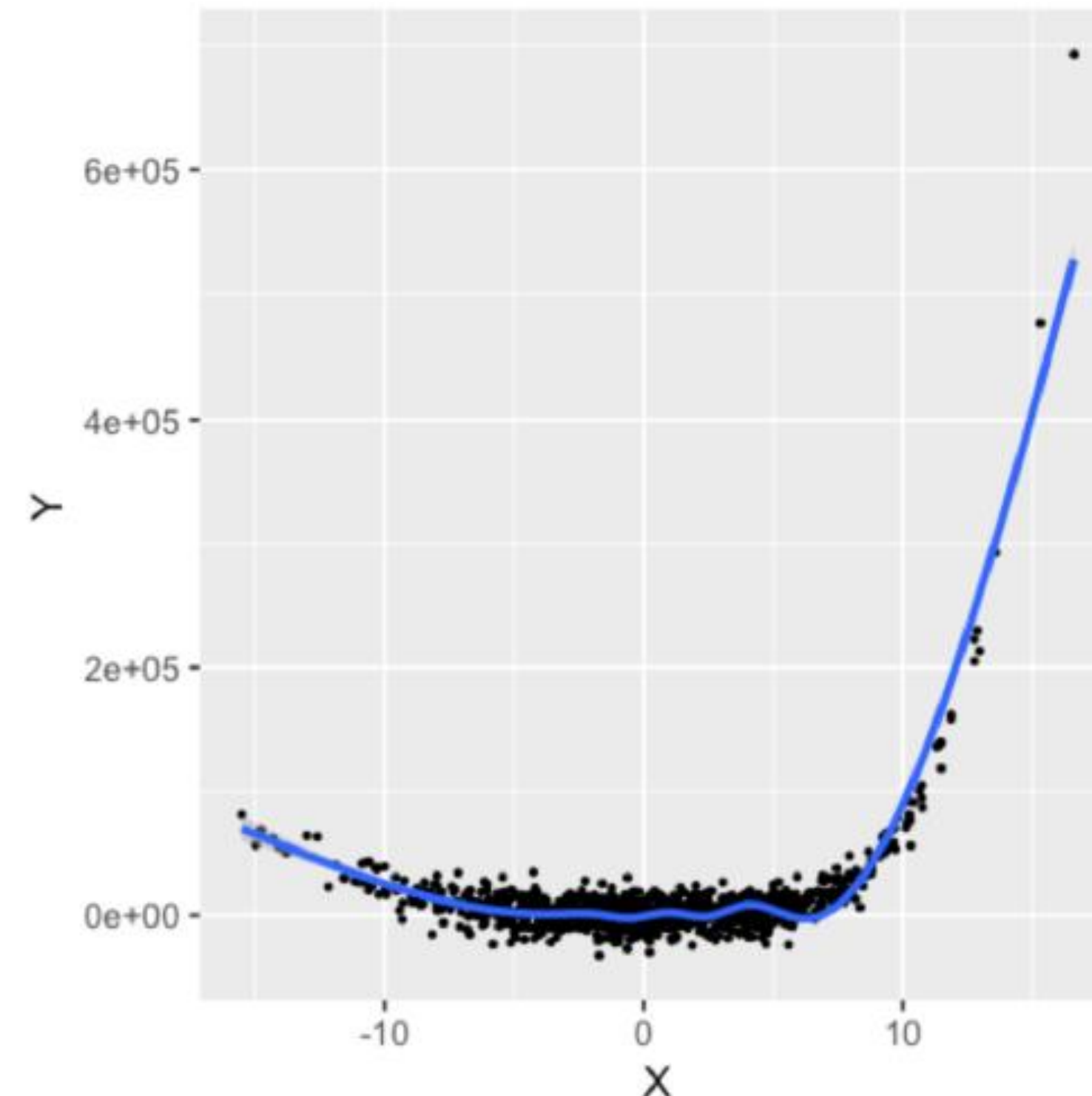


K^{th} degree polynomial models

- Expand x out to the power k .
- Include *all* terms up to k .

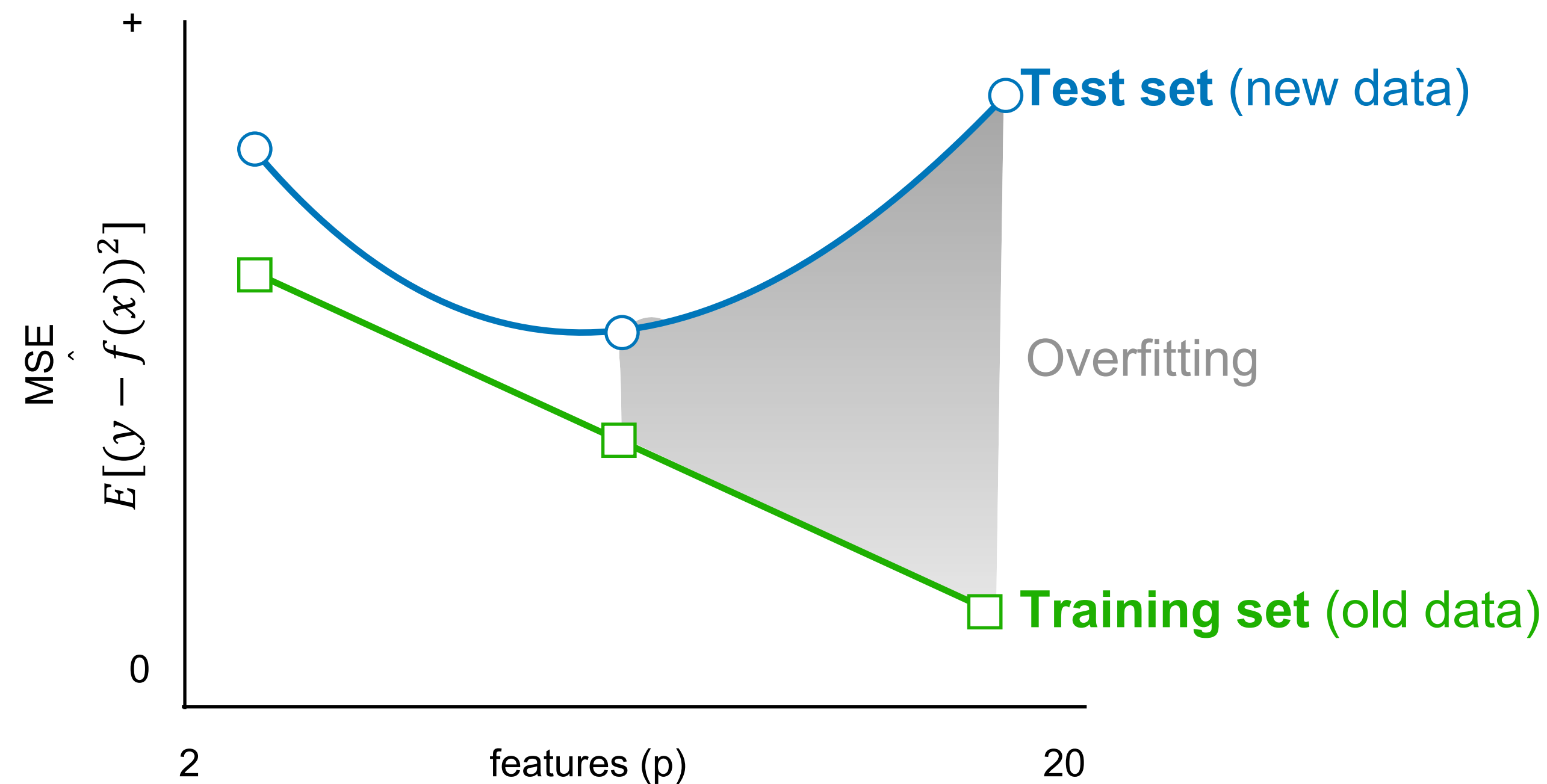
$$Y = \hat{\beta}_0 + \sum_{j=1}^K \hat{\beta}_j X^j$$
$$= \text{poly}(x, k)$$

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \beta_4 X^4 + \beta_5 X^5$$



Complexity

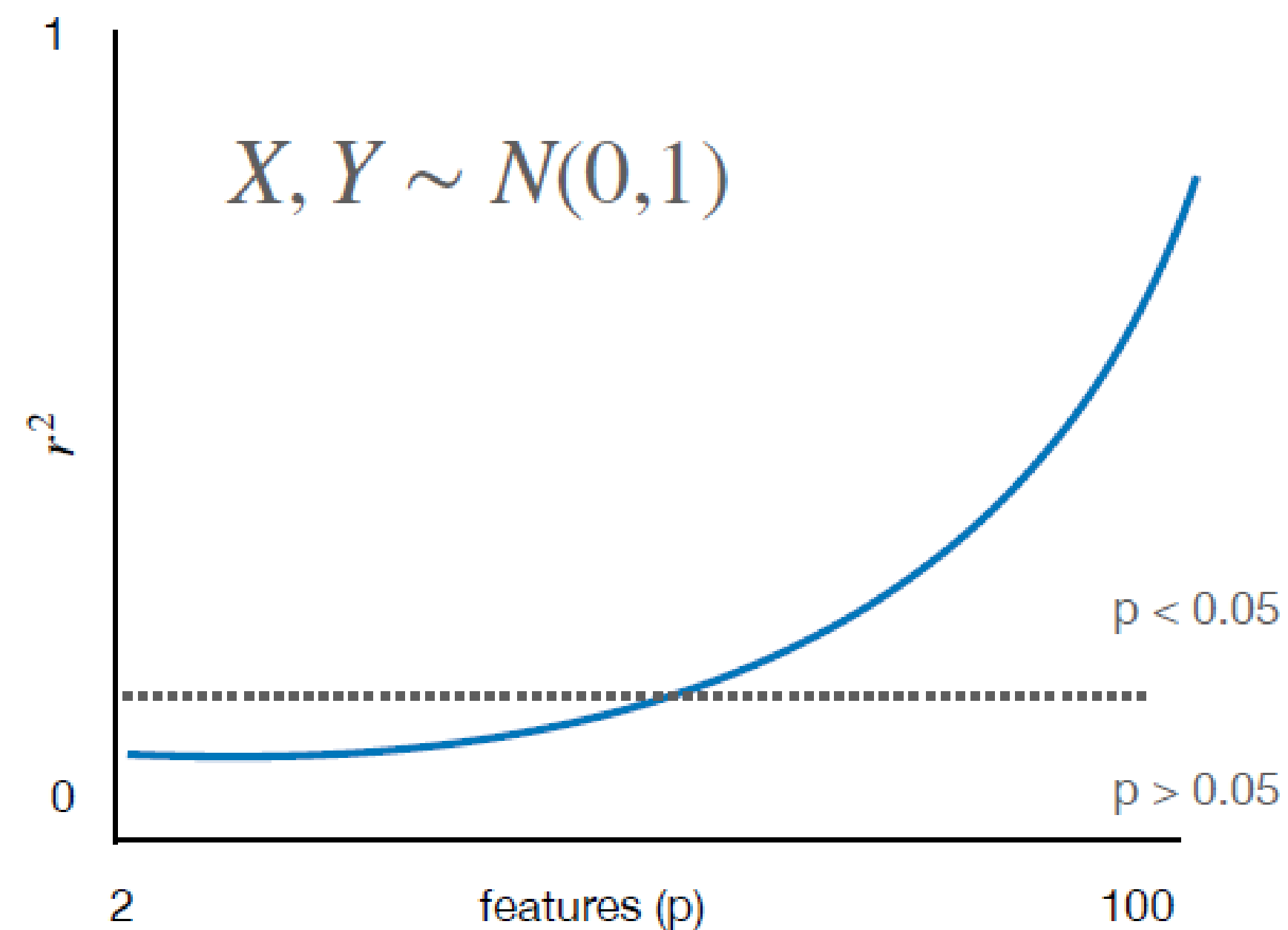
- p is the number of features (i.e., predictor variables) in X .
- $n \times p$ defines the flexibility/variance of a model.
- $k = p$ for k^{th} order polynomial models.



Risk to inference

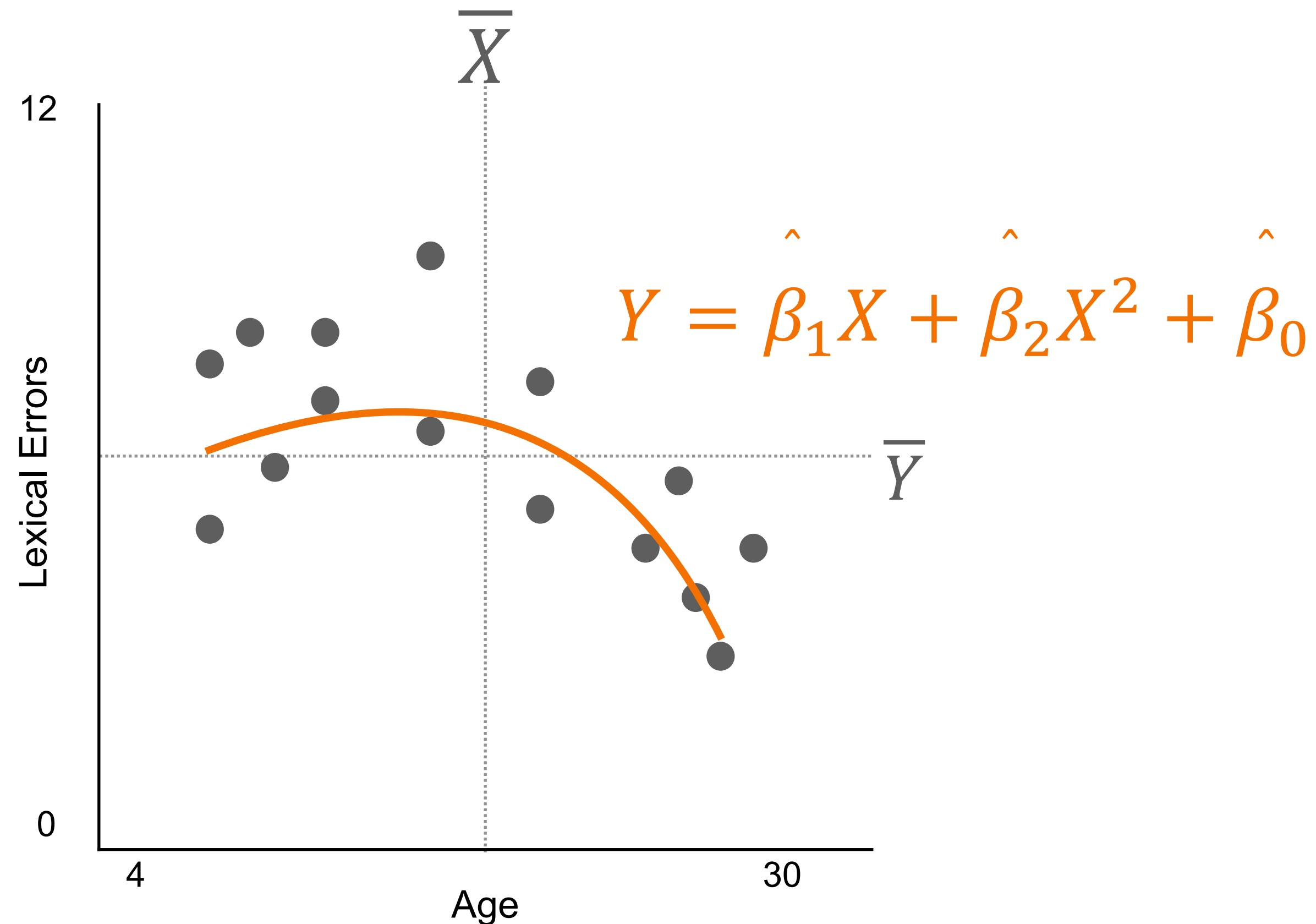
- Overfitting means that increasing p risks explaining a “statistically significant” portion of variance by chance.

$$Y = \hat{\beta}_0 + \sum_{j=1}^K \hat{\beta}_j X^j$$
$$= \text{poly}(x, k)$$



Risk to interpretation

Example



Transformation distort the interpretation of the underlying construct:

What is: Age^2 ?

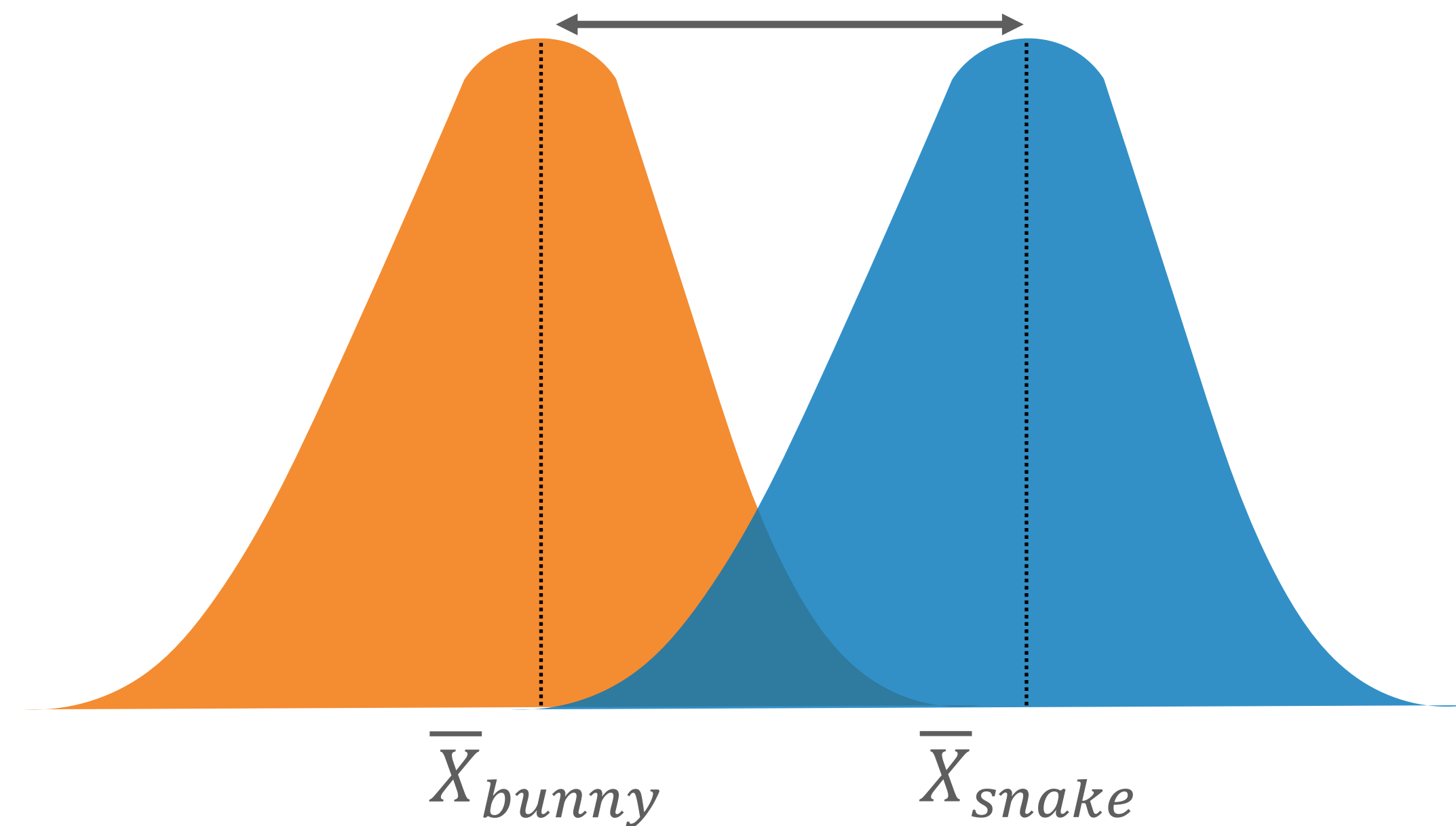
Isomorphisms with other statistical tests

The t-test

Q: Do infants attend more to pictures of snakes or bunnies?

Trial	Look time (ms)	Snake picture	Bunny picture
1	1600	1	0
2	120	0	1
3	874	1	0
4	333	0	1
5	740	1	0
6	201	0	1

$$t = \frac{\bar{X}_{snake} - \bar{X}_{bunny}}{\sigma_{snake,bunny}}$$



The t-test

Q: Do infants attend more to pictures of snakes or bunnies?

Trial	Look time (ms)	Snake picture	Bunny picture
1	1600	1	0
2	120	0	1
3	874	1	0
4	333	0	1
5	740	1	0
6	201	0	1

Regression form:

$$Y_{time} = \hat{\beta}_0 + \hat{\beta}_1 X_{snake} + \hat{\beta}_2 X_{bunny}$$

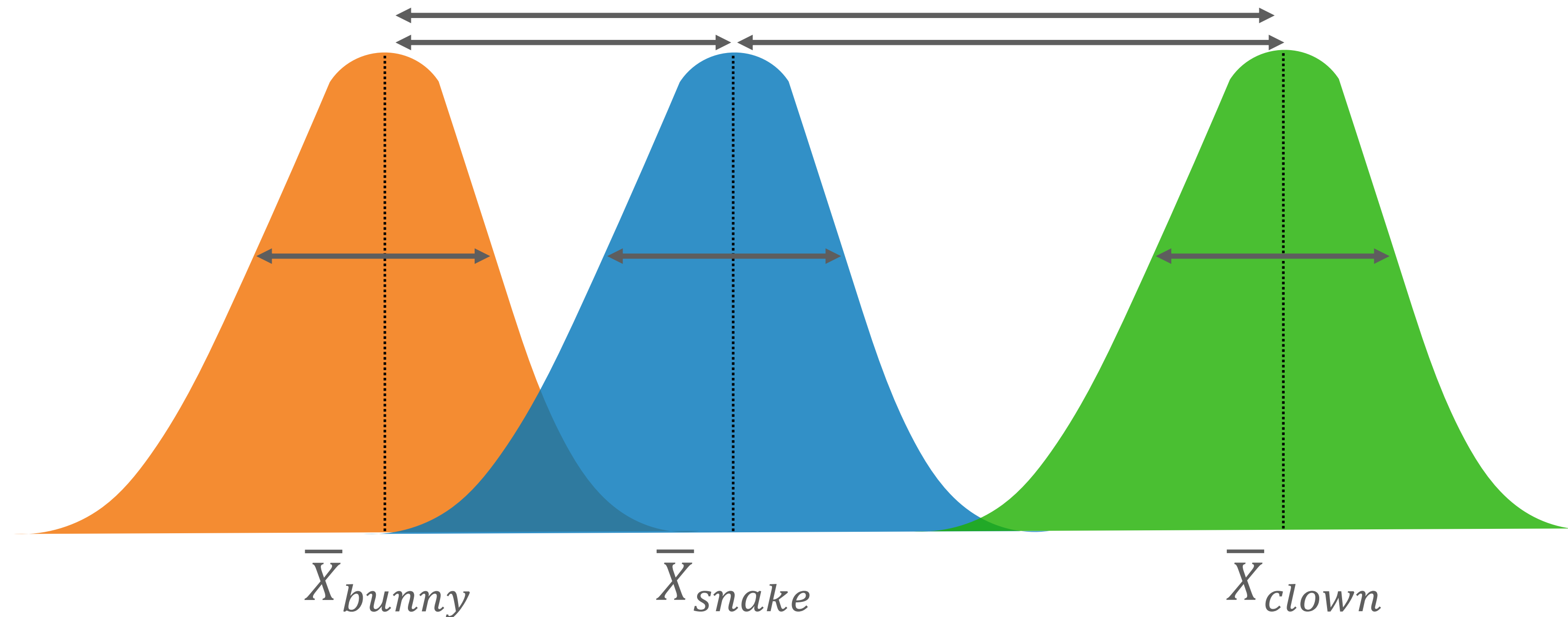
$$t = \frac{\hat{\beta}_1 - \hat{\beta}_2}{SE(\hat{\beta}_1 - \hat{\beta}_2)}$$

ANOVA

Q: Do infants attend more to pictures of snakes, bunnies, or clowns?

Trial	Look time (ms)	Snake picture	Bunny picture	Clown picture
1	1600	1	0	0
2	120	0	1	0
3	3010	0	0	1
4	333	0	1	0
5	740	1	0	0
6	2237	0	0	1

$$F = \frac{\sigma_{between}^2}{\sigma_{within}^2} = \frac{\frac{\sum_{i=1}^p n_i (y_i - \bar{y}_i)^2}{p - 1}}{\frac{\sum_{i=1}^p \sum_{j=1}^{n_i} (y_{i,j} - \bar{y}_i)^2}{n - p}}$$



ANOVA

Q: Do infants attend more to pictures of snakes, bunnies, or clowns?

Trial	Look time (ms)	Snake picture	Bunny picture	Clown picture
1	1600	1	0	0
2	120	0	1	0
3	3010	0	0	1
4	333	0	1	0
5	740	1	0	0
6	2237	0	0	1

Regression form:

$$Y_{time} = \hat{\beta}_0 + \hat{\beta}_1 X_{snake} + \hat{\beta}_2 X_{bunny} + \hat{\beta}_3 X_{clown}$$

$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$TSS = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$F = \frac{\frac{1}{p}(TSS - RSS)}{\frac{RSS}{n - p - 1}}$$

Take home message

- Linear regression models are extensible enough to test many questions typically evaluated using more narrow statistical methods.

Linear regression model: Model Assumptions

What **must be true** for the equation to be a reasonable description of the world?

Fundamental form:

$$Y = f(X) + \epsilon$$

$$= \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \epsilon$$

slope*X + intercept

Necessary Assumptions:

1. $E[Y|X] = f(X) \rightarrow$ for observations with the same predictors, the *expected (average)* outcome is the same.
2. $f(X) = \beta X \rightarrow$ the predictors combine additively to determine the expected outcome.
3. $\epsilon \rightarrow$ individual outcomes vary around the expected outcome.
4. $E[\epsilon | X] = 0 \rightarrow$ exogeneity: there aren't hidden variables simultaneously affecting X and Y .
5. $Cov(\epsilon_i, \epsilon_j) = 0$: iid observations $\rightarrow$ observations provide independent information

Linear Regression: Coefficient validity

If we fit the regression, will the slopes we estimate recover the **true relationship**?

Fundamental form:

$$Y = f(X) + \epsilon$$

$$= \underbrace{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}_{\text{slope} * X + \text{intercept}} + \epsilon$$

β estimates are unbiased when:

1. **Exogeneity:** there aren't hidden variables simultaneously affecting X and Y.
2. **Representative/independent sampling:** Your dataset is not systematically selected in a way that ties X to the error.
3. **No Multicollinearity:** No predictor is an exact linear combination of others.

Linear Regression: Inference validity

How **certain** are we of the estimates of the relationship?

Fundamental form:

$$\begin{aligned} Y &= f(X) + \epsilon \\ &= \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \epsilon \\ &\quad \underbrace{\hspace{10em}}_{\text{slope} * X + \text{intercept}} \end{aligned}$$

When SEs and p-values are correct:

1. **Homoskedasticity:** the spread of residuals is similar across levels of the predictors.
2. **Independent errors:** iid observations: observations provide independent information
3. **Normality:** residuals are approximately normal.

Linear Regression: Assumption salad

Fundamental form:

$$Y = f(X) + \epsilon$$

$$= \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \epsilon$$

slope*X + intercept

Model certainty: assumptions focus on data generation process: $X\beta + \epsilon$

Coefficient certainty: assumptions focus on estimation: $\hat{\beta}$

Inference certainty: assumptions focus on uncertainty: $SE(\hat{\beta})$